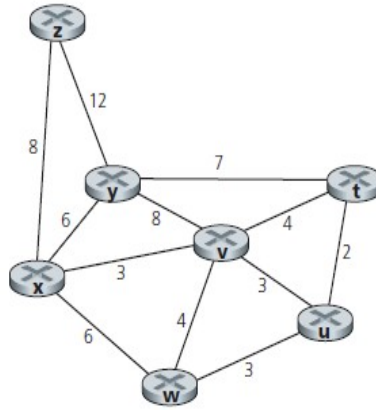
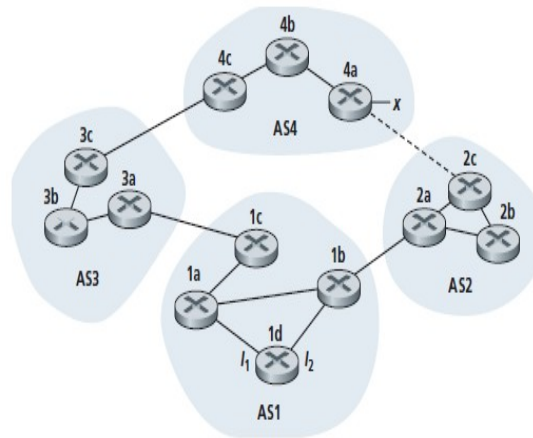


Q1: Consider the following network. With the indicated link costs, use Link State routing algorithm to compute the shortest path from t to all network nodes. Clearly show the table entries for computing the next node in the shortest path tree. Draw the final shortest path tree with all the costs.



Q2: Consider the network given below. Consider the network shown below. Suppose AS3 and AS2 are running OSPF for their intra-AS routing protocol. Suppose AS1 and AS4 are running RIP for their intra-AS routing protocol. Suppose eBGP and iBGP are used for the inter-AS routing protocol. Initially suppose there is *no* physical link between AS2 and AS4.



1. Router 3c learns about prefix x from which routing protocol: OSPF, RIP, eBGP, or iBGP?

Answer: _____

Reason:

2. Router 3a learns about x from which routing protocol?

Answer: _____

Reason:

3. Router 1c learns about x from which routing protocol?

Answer: _____

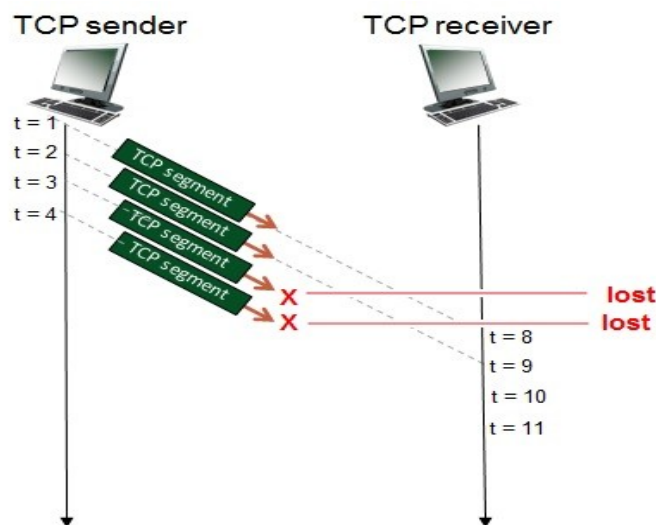
Reason:

4. Router 1d learns about x from which routing protocol?

Answer: _____

Reason:

1. Consider the figure below in which TCP a sender and receiver communicate over a connection in which the sender-to-receiver segments may be lost. The TCP sender sends initial window of four segments at $t=1,2,3,4$, respectively. Suppose the initial value of the sender-to-receiver sequence number is 96 and the first four segments *each* contain 566 bytes. The delay between the sender and the receiver is 7 time units, and so the first segment arrives at the receiver at $t=8$. As shown in the figure, two of the four segment(s) are lost between the sender and the receiver.



List the sequence of acknowledgements transmitted by the TCP receiver in response to the receipt of the segments actually received.

<u>Sender-to-Receiver</u>	<u>Time Segment Sent</u>	<u>Sender-to-Receiver segment sequence value</u>
Segment 1	1	96
Segment 2	2	662
Segment 3	3	1228
Segment 4	4	1794

CS307 (E)
Quiz 3 (Solution)
Fall 2019

0110011001100000
0101010101010101
1000111100001100

The sum of first two of these 16-bit words is

0110011001100000
0101010101010101
1011101110110101

Adding the third word to the above sum gives

1011101110110101
1000111100001100
0100101011000010

Note that this last addition had overflow, which was wrapped around. The 1s complement is obtained by converting all the 0s to 1s and converting all the 1s to 0s. Thus the 1s complement of the sum 0100101011000010 is 1011010100111101, which becomes the checksum. At the receiver, all four 16-bit words are added, including the checksum. If no errors are introduced into the packet, then clearly the sum at the receiver will be 1111111111111111. If one of the bits is a 0, then we know that errors have been introduced into the packet.

National University of Computer and Emerging Sciences, Lahore Campus
Quiz3 [BS(CS): Section C] Fall 2022

Computer Networks (Code: CS3001)

Quiz Date: October 24, 2022

Total Marks: 10

Duration: 15 -Minutes

Name ----- Roll #----- Section -----

Instructions: Attempt all questions on this sheet. You can make use of rough sheet (do not attach to this sheet).

Q1: Encircle the correct option(s):

(4 Marks)

CLO 1

(i) Length field of UDP segment indicates the length of -----.

A. UDP header B. UDP payload C. TCP header **D. None of these**

(ii) Urg data pointer field in TCP segment consists of _____ bits.

A. 8 **B. 16** C. 4 D. 1

(iii) Sequence number field in TCP segment consists of _____ byte(s).

A. 16 **B. 32** C. 8 D. 1

(iv) When out-of-order segment arrives, TCP receiver immediately -----.

A. discards it B. sends an ACK **C. sends duplicate ACK**

Q2: Host A and B are communicating over a TCP connection, and Host B has already received from A all bytes up through byte 126. Suppose Host A then sends two segments to Host B back-to-back. The first and second segments contain 80 and 40 bytes of data, respectively. Host B sends an acknowledgment whenever it receives a segment from Host A. In the first segment, the sequence number is 127, the source port number is 302, and the destination port number is 80. Host B sends an acknowledgment whenever it receives a segment from Host A. Show necessary working **(2+2+2 = 6 Marks)** **CLO 4**

- a. What is the sequence number and destination port number in the second segment sent from Host A to B?
 - b. If the first segment arrives before the second segment, then, in the acknowledgment of the first arriving segment by host B to A, what will be the acknowledgment number and the destination port number?
 - c. If the second segment arrives before the first segment, then, in the acknowledgment of this received segment, what will be the acknowledgment number and source port number?
-

Start writing your Answer to Q2 from here and then use backside of this sheet.

Q2 Solution:

a. Sequence number in second segment = Sequence number in first segment + 80 = 127 + 80 = 207

Destination port number= 80

b. ACK number= 207

Destination port number= 302

c. ACK number= 127

Destination port number= 80

National University of Computer and Emerging Sciences, Lahore Campus
Quiz3 [BS(CS): Section D] Fall 2022

Computer Networks (Code: CS3001)

Quiz Date: October 24, 2022

Total Marks: 10

Duration: 15 -Minutes

Name ----- Roll #----- Section -----

Instructions: Attempt all questions on this sheet. You can make use of rough sheet (do not attach to this sheet).

Q1: Encircle the correct option(s): (3 Marks) **CLO 1**

(i) Destination port number field in TCP segment consists of _____ bits.

- A. 8 **B. 16** C. 4 D. 1

(ii) Which of the following is a pipelined protocol?

- A. rdt 2.0 B. rdt 3.0 C. rdt 2.1 **D. None of these**

(iii) When out-of-order segment arrives, TCP receiver immediately -----.

- A. discards it B. sends an ACK **C. sends duplicate ACK**

Q2: Station A needs to send a message consisting of 9 packets (numbered as 1 to 9) to station B using a sliding window (window size 3) and Go-Back-N error control strategy. All packets are ready and immediately available for transmission. If every 5th transmitted packet gets lost (ACKs from B never get lost), then how many packets A will transmit for sending entire message to B? Moreover, write packet numbers which get lost during this process. Show necessary working, (7 Marks) **CLO 4**

Start writing your Answers to Q2 onward from here and then use backside of this sheet.

Q2 Solution:

Let packets are numbered as 1 to 9.

Window size = 3

Every 5th packet gets dropped

Step 1 view	3rd packet (3)	2 nd packet (number 2)	1 st packet (number 1)
Step 2 view (after ACK for pkt # 1)	4th packet (number 4)	3rd packet (number 3)	2 nd packet (number 2)
Step 3 view (after ACK for pkt # 2)	5th packet (number 5)	4th packet (number 4)	3rd packet (number 3)
Step 4 view (after ACK for pkt # 3)	6th packet (number 6)	5th packet (number 5)	4th packet (number 4)
Step 5 view (after ACK for pkt # 4)	7th packet (number 7)	6th packet (number 6)	5th packet (number 5)
Step 6 view (After loss of 5 th - pkt # 5)	10th packet (number 7)	9th packet (number 6)	8th packet (number 5)

Step 7 view (after ACK for pkt # 5)	11th packet (number 8)	10th packet (number 7)	9th packet (number 6)
Step 8 view (after ACK for pkt # 6)	12th packet (number 9)	11th packet (number 8)	10th packet (number 7)
Step 9 view (After loss of 10 th - pkt # 7)	15th packet (number 9)	14th packet (number 8)	13th packet (number 7)
Step 10 view (after ACK for pkt # 7)		15th packet (number 9)	14th packet (number 8)
Step 11 view (after ACK for pkt # 8)			15th packet (number 9)
Step 12 view (After loss of 15 th - pkt # 9)			16th packet (number 9)
Step 13 view (after ACK for pkt # 9)			

Total number of packets sent: 16

Packet numbers which get lost: 5 (5th packet), 7 (10th packet) and 9 (15th packet)

Note: Kindly verify that answers are provided after performing necessary steps (not necessarily as mentioned above, concept should be clear).

If just answers are given, then give maximum 4 marks for all correct answers (1 for each). 1 mark for each correctly marked packet number.

97

National University of Computer and Emerging Sciences, Lahore Campus
Quiz3 [BS(CS): Section D] Fall 2023

Computer Networks (Code: CS3001)

Quiz Date: November 2, 2023

Total Marks: 10

Duration: 20 -Minutes

Name

Roll #

Section

Instructions: Attempt all questions on this sheet. You can make use of rough sheet (do not attach to this sheet).

Q1: Host A and B are communicating over a TCP connection, and Host B has already received from A all bytes up through byte 126. Suppose Host A then sends two segments to Host B back-to-back. The first and second segments contain 80 and 40 bytes of data, respectively. In the first segment, the sequence number is 127, the source port number is 302, and the destination port number is 80. Host B sends an acknowledgment whenever it receives a segment from Host A.

(6 Marks)

- What is the sequence number and destination port number in the second segment sent from Host A to B?
- If the first segment arrives before the second segment, then, in the acknowledgment of the first arriving segment by host B to A, what will be the acknowledgment number and the destination port number?
- If the second segment arrives before the first segment, then, in the acknowledgment of this received segment, what will be the acknowledgment number and source port number?

Q2: Suppose you have the following 2 bytes: 11111010 and 01100101. Compute UDP checksum if UDP uses 8-bit words in computing the checksum. Also state how will the receiver know whether an error has occurred or not.

(4 marks)

Start writing your Answers from here and then use backside of this sheet.

Q.1 a)

Ans. Sequence number = ~~127~~ 207 ✓
Source port number = 302 ✓
Destination port number = 80 ✓

$$\begin{array}{r} 127 \\ + 80 \\ \hline 207 \end{array}$$

b)

Ans. Acknowledgement number = 207 ✓
Destination port number = 302 ✓

c)

Ans. Acknowledgement no = 127 ✓
Dest. port number = 302 ✓
Source port = ?

Q.2

Ans.

$$\begin{array}{r} \text{B} \quad 0 \quad 0 \\ 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \\ 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \\ \hline 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \\ 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\ 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \end{array} \quad 2$$

The receiver adds the two 8 bit words and the checksum. If any of the bits have a zero, an error has occurred. *2*

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2023
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	19-October-2023	Section	5J
	Exam Type:	Quiz 3 - Chapter 3	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

- Sequence number field in TCP segment consists of _____ byte(s).
 - 16
 - 32
 - 8
 - 1
- User datagram protocol is called connectionless because:
 - All UDP packets are treated independently by transport layer.
 - No connection is made before the transfer of actual data
 - Both a & b
 - None
- UDP provides
 - Flow Control
 - Congestion Control
 - Fast Delivery
 - None

True/False:

- Flow Control: do not overwhelm the network. [T / F]
- Multi-casting is possible in TCP. [T / F]

Q2. Suppose that TCP's current estimated values for the round-trip time (estimated RTT) and deviation in the RTT (Dev RTT) are 210 msec and 49 msec, respectively. Suppose that the next two measured values of the RTT are 280, 210.

Compute TCP's new value of estimated RTT, Dev RTT, and the TCP timeout value after each of these three measured RTT values is obtained. Use the values of $\alpha = 0.125$, $\beta = 0.25$.

[10 marks] [CLO 3]

Solution Quiz#3(5J)

MCQ's

1. b
2. c
3. c

T/F

1. F
2. F

Q2.

- After the first RTT estimate is made:
Estimated RTT = 218.75 msec
Dev RTT = 52.06 msec
Timeout Interval = 427 msec
- After the second RTT estimate is made:
Estimated RTT = 217.66 msec
Dev RTT = 40.96 msec
Timeout Interval = 381.50 msec

National University of Computer and Emerging Sciences, Lahore Campus

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	Program:	BS (Computer Science)	Semester:	Fall 2023
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	19-October-2023	Section	5K
	Exam Type:	Quiz 3 - Chapter 3	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

1. When out-of-order segment arrives, TCP receiver immediately -----.
 - a) Discards it
 - b) Sends an ACK
 - c) sends duplicate ACK
 - d) None of the above

2. User datagram protocol is called connectionless because:
 - a) All UDP packets are treated independently by transport layer.
 - b) No connection is made before the transfer of actual data
 - c) a & b
 - d) None

3. Transport layer aggregates data from different applications into a single stream before passing it to
 - a) Network layer
 - b) Transport layer
 - c) Data link layer
 - d) Application layer

True/False:

1. TCP performs a fast retransmit upon receiving a total of three (3) same ACKs. [T / F]
2. Multi-casting is not possible in UDP. [T / F]

Q2. Fill in the incomplete:

[4] [CLO 3]

<i>Properties</i>	<i>Stop and Wait</i>	<i>GO BACK N</i>	<i>SELECTIVE REPEAT</i>
Sender Window Size	1		
Reciever Window Size			N

Q3. Host A and B are communicating over a TCP connection, and Host B has already received from A all bytes up through byte 126. Suppose Host A then sends two segments to Host B back-to-back. The first and second segments contain 80 and 40 bytes of data, respectively. In the first segment, the sequence number is 127, the source port number is 302, and the destination port number is 80. Host B sends an acknowledgment whenever it receives a segment from Host A.
[3 + 3 = 6 Marks] [CLO 3]

- (a) What is the sequence number and destination port number in the second segment sent from Host A to B?
- (b) If the first segment arrives before the second segment, then, in the acknowledgment of the first arriving segment by host B to A, what will be the acknowledgment number and the destination port number?

Solution Quiz#3(5K)

MCQ's

1. c
2. c
3. a

T/F

1. F
2. F

Q2. Solution:-

<i>Properties</i>	<i>Stop and Wait</i>	<i>GO BACK N</i>	<i>SELECTIVE REPEAT</i>
Sender Window Size	1	N	N
Receiver Window Size	1	1	N

Q3. Solution:-

(a) Sequence number in second segment = Sequence number in second segment + 80
= 127 + 80 = 207

Destination port number = 80

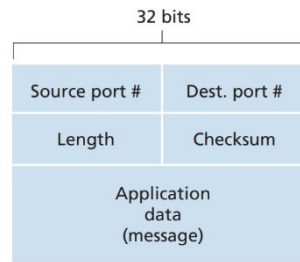
(b) ACK number = 207

Destination port number = 302

COMPUTER NETWORKS QUIZ-03 (B) SOLUTION – BCS-5G

1.

- a. First segment: seq= 494 , bytes= 499
second segment: seq= 993, bytes=499
third segment: seq= 1492, bytes 499
- b. ACK for 2nd segment: 993
ACK for 3rd segment: 993



- c. **UDP segment structure**
- d. **Size of UDP datagram:**

UDP Header + application layer header + message data

= 8 bytes + 700 bytes + 8 bytes

= 716 bytes

2. **EstimatedRTT = (1- α)*EstimatedRTT + α*SampleRTT**
TimeoutInterval = EstimatedRTT + 4*DevRTT
DevRTT = (1-β)*DevRTT + β*|SampleRTT-EstimatedRTT|

- a. **Sample 1:**

EstimatedRTT = (1-0.125)(200) + (0.125)(200) = 202.5

DevRTT = (1-0.25)(20) + 0.25(|220-200|) = 20

TimeoutInterval = 202.5 + 4(20) = 282.5

- b. **Sample 2:**

EstimatedRTT = (1-0.125)(202.5) + (0.125)(210) = 203.44

DevRTT = (1-0.25)(20) + (0.25)(|210-202.5|) = 16.875

TimeoutInterval = 203.44 + 4(16.875) = 270.94

- c. **Sample 3:**

EstimatedRTT = (1-0.125)(203.44) + (0.125)(240) = 208.01

DevRTT = (1-0.25)(16.875) + (0.25)(|240-203.44|) = 21.796

TimeoutInterval = 208.01 + 4(21.796) = 295.194

3.

a. **Issue with Two-Way Handshake:**

Old or Delayed Packets: In a two-way handshake, there is no verification if the connection request is new or an outdated packet. A delayed packet could unintentionally trigger a connection, causing confusion.

Synchronization Problem: The two-way handshake does not confirm if both the client and server have successfully synchronized their sequence numbers, leading to possible data misalignment.

b. **Solution with Three-Way Handshake**

1. SYN (Synchronize): The client sends a connection request (SYN) to the server with an initial sequence number.

2. SYN-ACK (Acknowledge + Synchronize): The server responds with a SYN-ACK, acknowledging the client's SYN and sending its own initial sequence number.

3. ACK (Acknowledge): The client sends an ACK back to the server, confirming the server's sequence number.

This three-step process ensures:

Reliability: Both the client and server confirm the connection request, reducing the risk of old or duplicate connection attempts.

Synchronization: Both the client and server synchronize their sequence numbers, ensuring data is properly aligned for transmission.

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	07-October-2025	Section	5A , 5C
	Exam Type:	Quiz 3 - Chapter 3	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

- Which of the following is false with respect to TCP?
 - Connection Oriented
 - Process to Process
 - Transport Layer Protocol
 - Unreliable
- What is the header size of a UDP packet?
 - 8 Bytes
 - 8 Bits
 - 16 Bytes
 - 20 Bytes
- Which TCP mechanism is responsible for preventing a fast sender from overwhelming a slow receiver?
 - Congestion Control
 - Slow Start
 - Congestion Avoidance
 - Flow Control

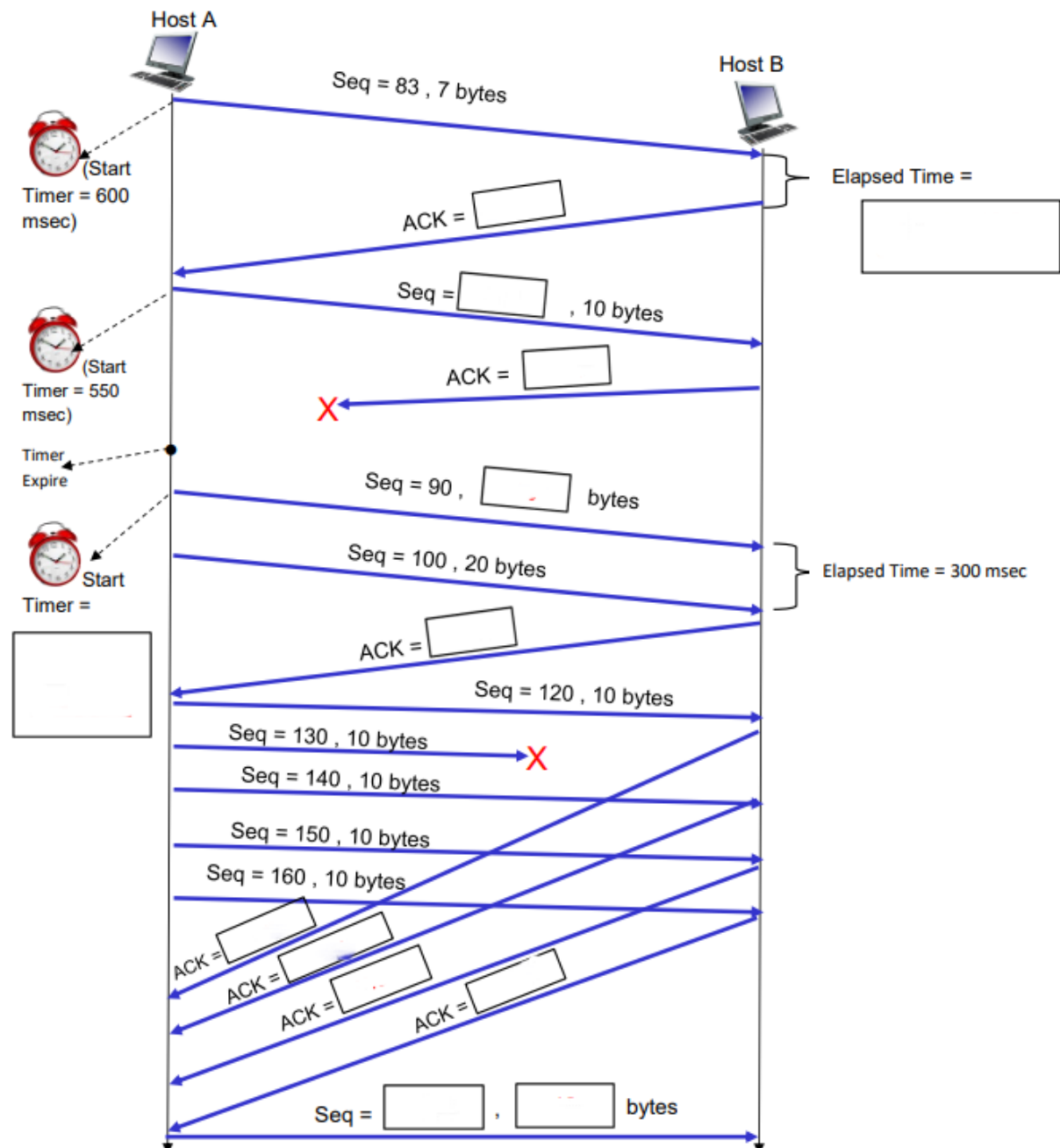
True/False:

- Selective Repeat makes better use of Bandwidth than Go-Back-N. [T / F]
- Selective Repeat uses cumulative ACK's. [T / F]

Q2:

[10 Marks] [CLO 3]

Refer to the TCP segments exchange between Host A to Host B. Fill in the missing values in the space provided in boxes:



**National University of Computer and Emerging Sciences, Lahore
Campus**

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Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

1. The _____ field is used to detect errors over the entire UDP packet.
 - a) Header
 - b) Checksum
 - c) Source Port
 - d) Destination Port
2. The mechanism in TCP with which the receiver of the data controls the amount of data that is to be sent by the sender is referred to as _____.
 - a) Flow Control
 - b) Error Control
 - c) Congestion Control
 - d) Error Detection
3. What is the primary difference between Go-Back-N and Selective Repeat protocols?
 - a) Go-Back-N uses cumulative acknowledgments, while Selective Repeat uses individual acknowledgments.
 - b) Go-Back-N has a larger window size compared to Selective Repeat.
 - c) Selective Repeat has a higher probability of packet loss than Go-Back-N.
 - d) Go-Back-N allows the receiver to request retransmission of specific lost packets, while Selective Repeat retransmits the entire window.

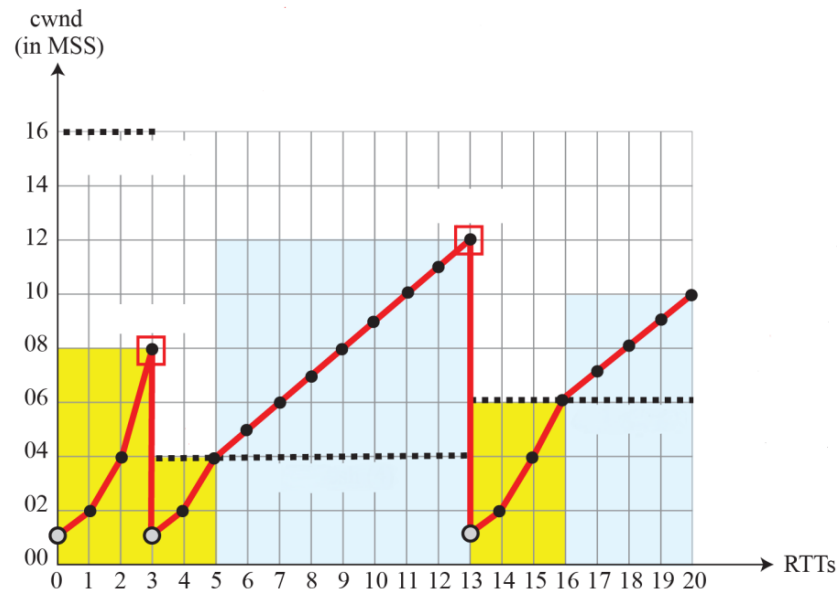
True/False:

1. Selective Repeat sender window size is 1. [T / F]
2. Selective Repeat is simpler to implement than Go-Back-N. [T / F]

Q2:

[10 Marks] [CLO 3]

Refer to Diagram below and answer the following questions:



- (1) The diagram shows which TCP (Reno or Tahoe)?
- (2) What happened at transmission round 3 (i.e. at RTT = 3), i.e. state the reason / event?
- (3) What happened at transmission round 5 (i.e. at RTT = 5), i.e. state the reason / event?
- (4) What happened at transmission round 13 (i.e. at RTT = 13), i.e. state the reason / event?
- (5) What happened at transmission round 16 (i.e. at RTT = 16), i.e. state the reason / event?
- (6) Which state is the TCP in between transmission rounds (RTT):
 - (a) 0 & 3
 - (b) 3 & 5
 - (c) 5 & 13
 - (d) 13 & 16
 - (e) 16 & 20

SOLUTION QUIZ-03 BCS-5A

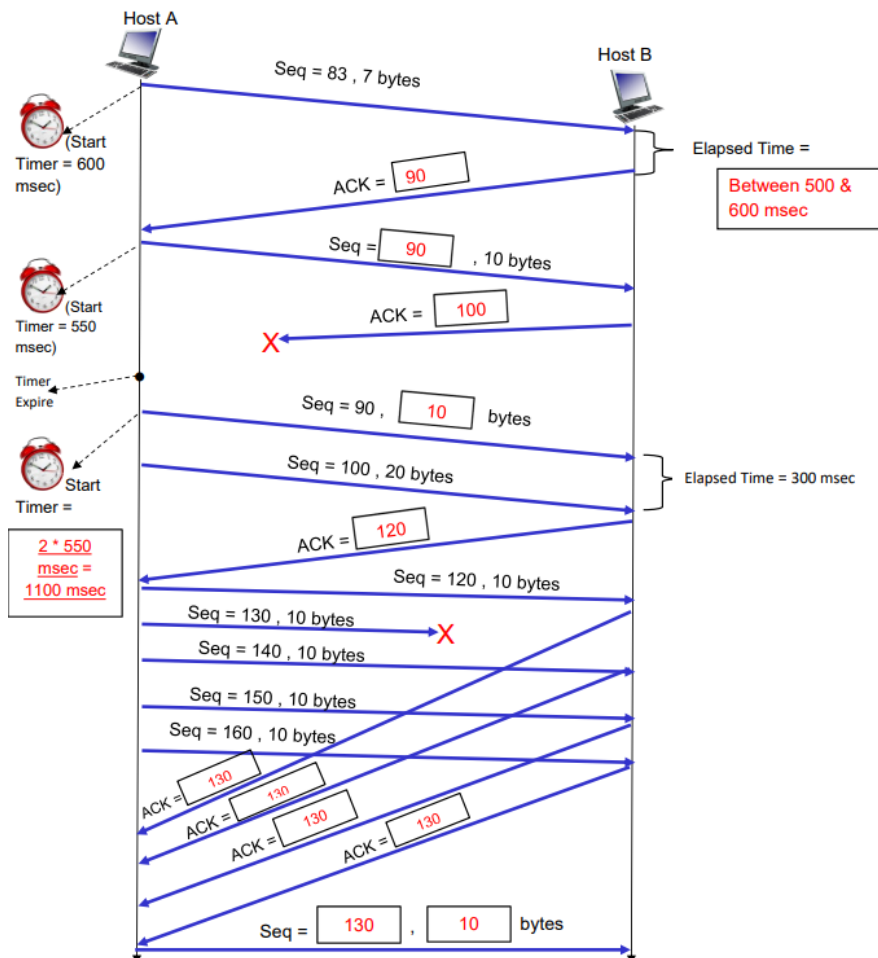
QUESTION-01

1. D
2. A
3. D

TRUE/FALSE

- i. True
- ii. False

QUESTION-02



SOLUTION QUIZ-03 BCS-5C

QUESTION-01

1. B
2. A
3. A

TRUE/FALSE

- i. False
- ii. False

QUESTION-02

1. Tahoe
2. Time-Out
3. ssth-reached
4. 3 duplicate ACK's received
5. Ssth-reached
6. Following are the Answers
 - a. Slow Start
 - b. Slow Start
 - c. Congestion Avoidance
 - d. Slow Start
 - e. Congestion Avoidance

SOLUTION QUIZ-03 BCS-6A

QUESTION-01

- 1) B
- 2) A
- 3) A

TRUE/FALSE

- i. False
- ii. True

QUESTION-02

- a) Sequence Number = 100
- b) ACK Number = 100
- c) Sequence Number = 200
- d) ACK Number = 300
- e) Fast Retransmit as 3 duplicate ACKs received.

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2024
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	19-March-2024	Section	6A
	Exam Type:	Quiz 3 - Chapter 3	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

- The _____ field is used to detect errors over the entire UDP packet.
 - UDP header
 - checksum
 - source port
 - destination port
- The mechanism in TCP with which the receiver of the data controls the amount of data that are to be sent by the sender is referred to as _____.
 - Flow control
 - Error control
 - Congestion control
 - Error detection
- What is the primary difference between Go-Back-N and Selective Repeat protocols?
 - Go-Back-N uses cumulative acknowledgments, while Selective Repeat uses individual acknowledgments.
 - Go-Back-N has a larger window size compared to Selective Repeat.
 - Selective Repeat has a higher probability of packet loss than Go-Back-N.
 - Go-Back-N allows the receiver to request retransmission of specific lost packets, while Selective Repeat retransmits the entire window.

True/False:

- TCP is a connectionless protocol. [T / F]
- UDP header size is 8 bytes? [T / F]

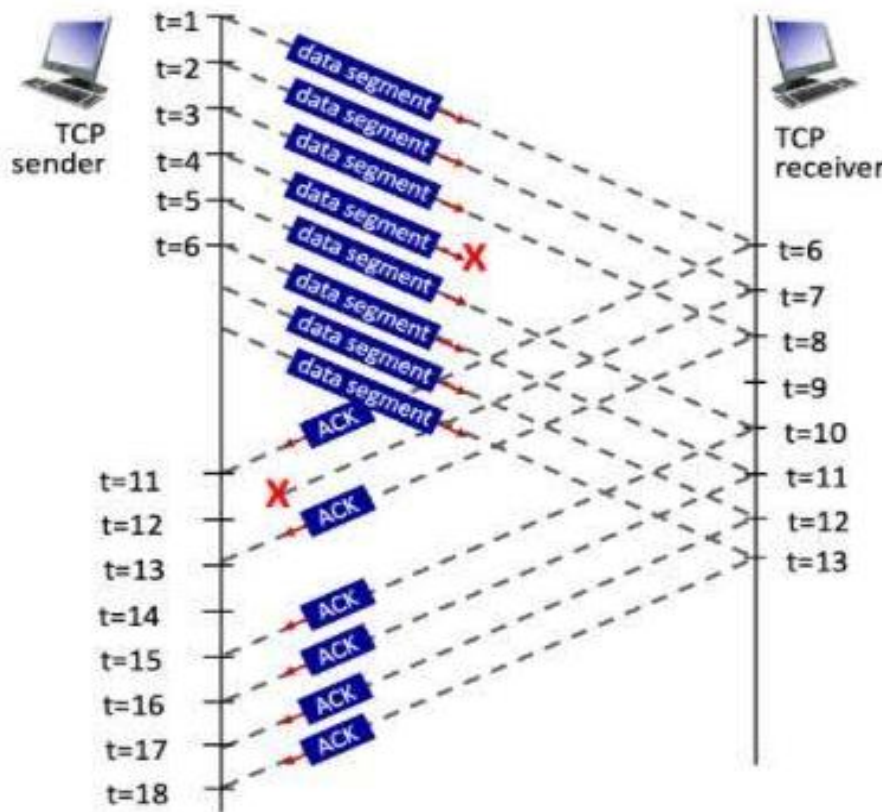
Question 2:

Consider the figure below, where a TCP sender sends 8 TCP segments at $t = 1, 2, 3, 4, 5, 6, 7, 8$. Suppose the initial value of the sequence number is 0 and every segment sent to the receiver **each contains 100 bytes**. The delay between the sender and receiver is 5 time units, and so the first segment arrives at the receiver at $t = 6$. The ACKs sent by the receiver at $t = 6, 7, 8, 10, 11, 12$ are shown. The

TCP segments (if any) sent by the sender at $t = 11, 13, 15, 16, 17, 18$ are not shown.

The segment sent at $t=4$ is lost, as is the ACK segment sent at $t=7$.

[2 + 2 + 2 + 2 + 2 = 10 Marks] [CLO 3]



- What is the sequence number of segment sent at $t=2$?
- What is the ACK value carried in the receiver to sender ACK sent at $t=6$?
- What is the sequence number of segment sent at $t=3$?
- What is the ACK value carried in the receiver to sender ACK sent at $t=10$?
- What does the sender do at $t=17$? (Name the Event and Give Reason)

SOLUTION QUIZ-03 BSE-6A

QUESTION-01

- 1) D
- 2) A
- 3) D

TRUE/FALSE

- i. False
- ii. True

QUESTION-02

- a) ACK1= 110
ACK2= 110
ACK3= 110
ACK4= 110
SEQ1= 110
- b) 20 bytes retransmitted
3 duplicate Acks caused **Fast Retransmit**

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2024
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	19-March-2024	Section	BSE-6A
	Exam Type:	Quiz 3 - Chapter 3	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

1. Which of the following is false with respect to TCP?
 - a) Connection-oriented
 - b) Process-to-process
 - c) Transport layer protocol
 - d) Unreliable

2. What is the header size of a UDP packet?
 - a) 8 bytes
 - b) 8 bits
 - c) 16 bytes
 - d) 20 bytes

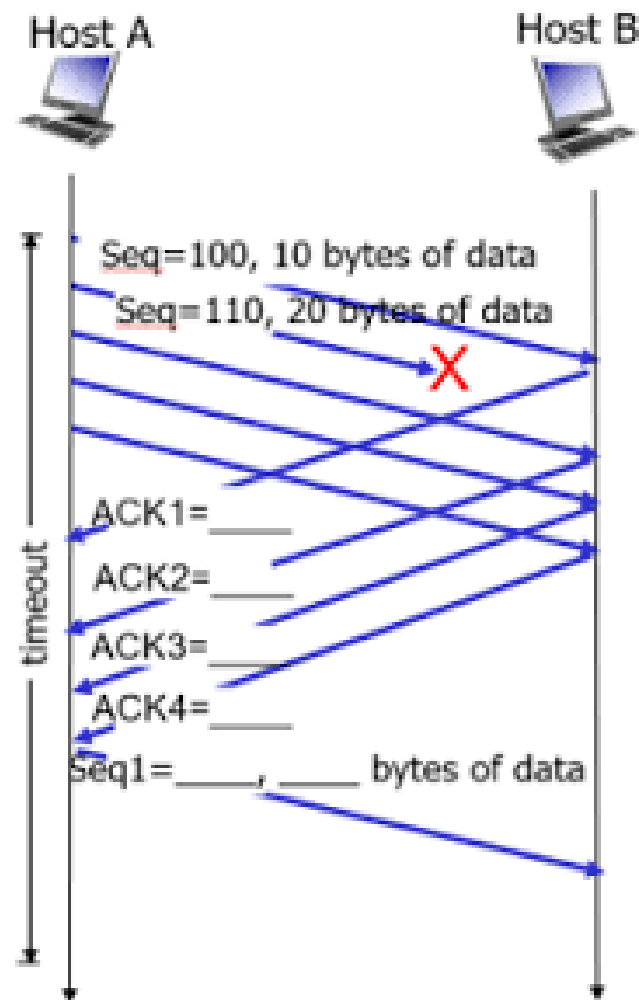
3. Which TCP mechanism is responsible for preventing a fast sender from overwhelming a slow receiver?
 - a) Congestion Control
 - b) Slow start
 - c) Congestion avoidance
 - d) Flow control

True/False:

- i. Selective Repeat uses cumulative acks [T / F]
- ii. UDP is frequently utilized in situations where minimizing delay is crucial, and a certain level of data loss is acceptable. [T / F]

Question 2

Look at the network communication scenario using TCP fast as shown in the following figure. The first packet Seq=100 arrives at the B successfully, but the second packet Seq = 110 with length 20 bytes was lost. [8 + 2 = 10 Marks] [CLO 3]



- What are the values of ACK1, ACK2, ACK3, ACK4 and SEQ1 respectively? Fill in the blanks in the diagram.
- How many bytes are transmitted in the final segment shown in diagram. Name the event that caused the transmission.

SOLUTION QUIZ-03 BCS-5C

QUESTION-01

1. b
2. c
3. c

TRUE/FALSE

- i. False
- ii. False

QUESTION-02

- After the first RTT estimate is made: Estimated RTT = 218.75 msec Dev RTT = 52.06 msec Timeout Interval = 427 msec
- After the second RTT estimate is made: Estimated RTT = 217.66 msec Dev RTT = 40.96 msec Timeout Interval = 381.50 msec

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	Program:	BS (Computer Science)	Semester:	Spring 2025
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	Exam Type:	Quiz 3 - Chapter 3	Page(s):	2

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Q1. Encircle the correct option:

[5 marks] [CLO 3]

- When out-of-order segment arrives, TCP receiver immediately -----.
 - Discards it
 - Sends a duplicate ACK
 - Sends an ACK
 - None of the above
- User datagram protocol is called connectionless because:
 - All UDP packets are treated independently by transport layer.
 - No connection is made before the transfer of actual data
 - a & b
 - None
- Transport layer aggregates data from different applications into a single stream before passing it to
 - Data link layer
 - Physical layer
 - Network layer
 - Application layer

True/False:

- TCP is a connectionless protocol. [T / F]
- Checksum with wraparound is used for **error correction** [T / F]

Question 2:**[Marks 10] [CLO 3]**

A network engineer is monitoring a TCP connection between a client and a remote server. To ensure smooth data transfer, TCP adjusts its timeout interval based on real-time network conditions. At the moment, TCP's Estimated RTT is 210 milliseconds, and the Deviation in RTT (Dev RTT) is 49 milliseconds. As new data packets are transmitted, the engineer records two new RTT measurements: 280 ms and 210 ms. Since TCP dynamically updates its timeout values to adapt to changing network conditions, the engineer needs to recalculate the following after each new RTT measurement:

1. The updated Estimated RTT
2. The updated Deviation RTT (Dev RTT)
3. The new TCP Timeout Interval

Using $\alpha = 0.125$ and $\beta = 0.25$, determine how these values change after each measurement. Consider how these adjustments impact the connection's reliability and responsiveness.

SOLUTION QUIZ-03 BSE-6C

QUESTION-01

1. b
2. c
3. c

TRUE/FALSE

- i. False
- ii. False

QUESTION-02

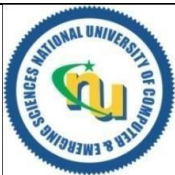
Q2. Solution:-

<i>Properties</i>	<i>Stop and Wait</i>	<i>GO BACK N</i>	<i>SELECTIVE REPEAT</i>
Sender Window Size	1	N	N
Receiver Window Size	1	1	N

Q3. Solution:-

- (a) Sequence number in second segment = Sequence number in second segment + 80
= $127 + 80 = 207$
Destination port number = 80
- (b) ACK number = 207
Destination port number = 302

**National University of Computer and Emerging Sciences, Lahore
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Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

1. Sequence number field in TCP segment consists of _____ bit(s).
 - a) 16
 - b) 32
 - c) 8
 - d) 1
2. User datagram protocol is called connectionless because:
 - a) All UDP packets are treated independently by transport layer.
 - b) No connection is made before the transfer of actual data
 - c) Both a & b
 - d) None
3. UDP provides
 - a) Flow Control
 - b) Congestion Control
 - c) Fast Delivery
 - d) None

True/False:

1. Flow Control: do not overwhelm the network. [T / F]
2. Multi-casting is possible in TCP. [T / F]

Question 2: Fill in the incomplete:

[4 Marks] [CLO 3]

<i>Properties</i>	<i>Stop and Wait</i>	<i>GO BACK N</i>	<i>SELECTIVE REPEAT</i>
Sender Window Size	1		
Receiver Window Size			N

Question 3. Host A and B are communicating over a TCP connection, and Host B has already received from A all bytes up through byte 126. Suppose Host A then sends two segments to Host B back-to-back. The first and second segments contain 80 and 40 bytes of data, respectively. In the first segment, the sequence number is 127, the source port number is 302, and the destination port number is 80. Host B sends an acknowledgment whenever it receives a segment from Host A. **[3 + 3 Marks] [CLO 3]**

(a) What is the sequence number and destination port number in the second segment sent from Host A to B?

(b) If the first segment arrives before the second segment, then, in the acknowledgment of the first arriving segment by host B to A, what will be the acknowledgment number and the destination port number?

SOLUTION QUIZ-03 BSE-6A

QUESTION-01

1. **Handshake:**

- A -> B: SYN=1, Seq=1200
- B -> A: SYN=1, ACK=1, Seq=4800, Ack=1201
- A -> B: SYN=0, ACK=1, Seq=1201, Ack=4801 (First data can piggyback here, usually considered the 3rd step).

2. **Data Segments:**

- Seg 1: Seq=1201 (Bytes 1201-2200).
- Seg 2: Seq=2201 (Bytes 2201-3200).

3. **Lost Segment 1:**

- If Seg 1 is delayed but Seg 2 arrives, TCP sends a **duplicate ACK** for the *expected* byte.
- **Ack = 1201.** (It acknowledges receipt of everything *up to* 1201, signaling it is still waiting for 1201).

QUESTION-02

(b) **Sequence number of lost segment: 700**

Table 1 – Complete Solution

TXNR (RTT)	Recv Window (rwnd MSS)	Cong.W window (cwnd MSS)	ssthresh (MSS)	RecvACK No.	Seq No(s) Transmitted	Timer (ms)	No Loss Or Timeout Or 3 dupACK	TCP Mode (SS/CA/FR)
0	8	1	8	N/A	N/A	600	No Loss	N/A
1	8	1	8	N/A	0	600	No Loss	SS
2	8	2	8	100	100, 200	600	No Loss	SS
3	8	4	8	300	300, 400, 500, 600	600	No Loss	SS
4	8	8	8	700	700, 800, 900, 1000, 1100, 1200, 1300, 1400	600	3dupACK	SS
5	8	7	4	700 (dup)	700 (retx)	600	3dupACK	FR
6	8	4	4	800	800, 900, 1000, 1100	600	No Loss	CA
7	8	5	4	1200	1200, 1300, 1400, 1500, 1600	600	No Loss	CA

**National University of Computer and Emerging Sciences,
Lahore Campus**

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	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	20 minutes	Total Marks:	15
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Section:

Q1. Host A establishes a TCP connection with Host B using the three-way handshake. Host A selects an initial sequence number (ISN) of 1200, while Host B selects an ISN of 4800. Immediately after the connection is established, Host A requests to send 5000 bytes of data to Host B. Given that the Maximum Segment Size (MSS) is 1000 bytes: **[3+1+1 = 5 Marks] [CLO 2]**

- (a) Draw the 3-Way Handshake diagram, labeling all SYN bits, ACK bits, Seq numbers, and Ack numbers.

- (b) Show the first two data segments sent by Host A. What are their Sequence Numbers?

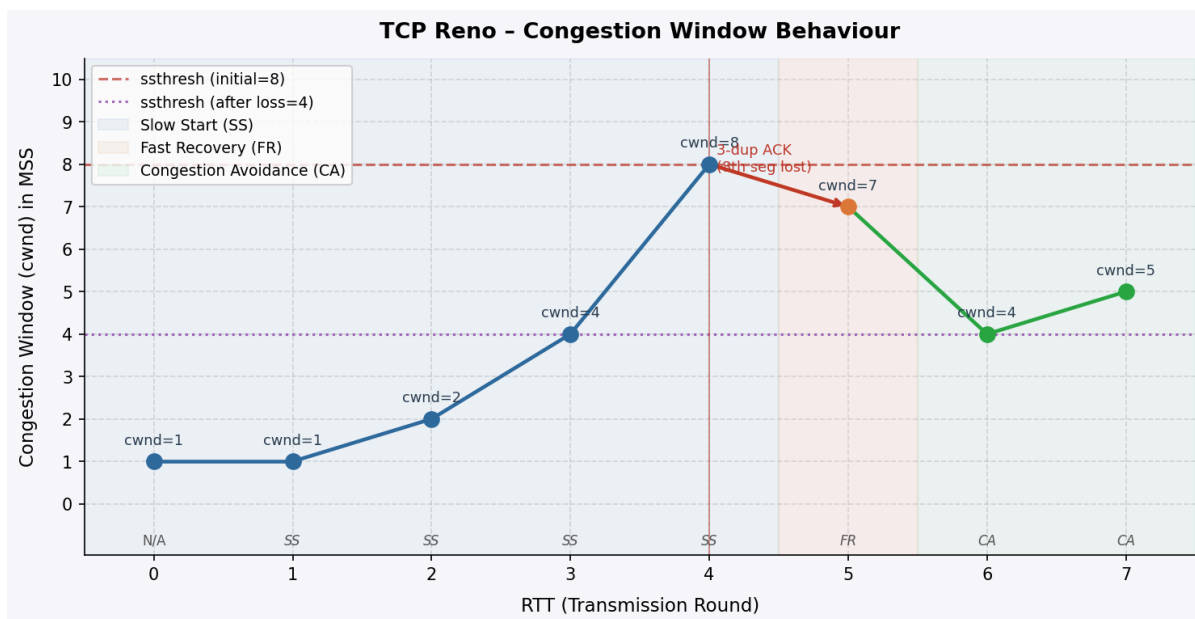
- (c) If the second data segment is received but the first is delayed, what Ack number will Host B send in response to the second segment?

Q2. Consider a TCP Reno connection as illustrated by the diagram below. Note the following:

- 1 MSS = 100 bytes
- At RTT = 0, the TCP connection has been established but no segment has been sent. Initial values are given in the table below.
- The sender starts transmission (first segment, seq. no. = 0) at RTT = 1.
- The 8th segment sent by the sender is lost.
- Out-of-sequence received segments are discarded (not buffered).
- SS = Slow Start, CA = Congestion Avoidance, FR = Fast Recovery, N/A = Not Applicable.
- Loss detected via 3 Duplicate ACKs.

(a) Fill in all missing entries in the TCP Sender Table (Table 1) below. Some entries are pre-filled for guidance.

(b) What is the sequence number of the segment that was lost? _____



TXNR(RTT)	rwnd (MSS)	cwnd (MSS)	ssthresh (MSS)	RecvACKNo.	Seq No(s) Transmitted	Timer (ms)	No LossOrTimeoutOr3dupACK	TCPMode(SS/CA/FR)
0	8	1	8	N/A	N/A	600	No Loss	N/A
1	8			N/A		600	No Loss	SS
2	8					600		
3	8	4				600	No Loss	
4	8		8	700		600		
5	8				700	600		
6	8					600	No Loss	
7	8	5				600	No Loss	CA

SOLUTION QUIZ-03 BSE-6B

QUESTION-01

a.

$$D_{trans} = \frac{L}{R} = \frac{4000}{100,000,000} = 0.00004 \text{ seconds} = 0.04 \text{ ms}$$

$$U_{sender} = \frac{D_{trans}}{D_{trans} + RTT}$$

$$U_{sender} = \frac{0.04}{0.04 + 10} = \frac{0.04}{10.04} \approx 0.003984$$

b.

To achieve 100% utilization, the sender must continuously transmit packets during the entire time it takes to get the first acknowledgment back (which is one full round trip plus the transmission time of the first packet).

We can calculate the required window size (W) by dividing the total time of one cycle by the transmission time of a single packet:

$$W = \frac{D_{trans} + RTT}{D_{trans}}$$

$$W = \frac{10.04 \text{ ms}}{0.04 \text{ ms}} = 251$$

Answer: A minimum window size of **251 packets** must be kept in flight to achieve 100% utilization.

QUESTION-02

Total number of packets (including both transmissions and re-transmissions) sent by host A is 16 as shown in the following timing diagram:

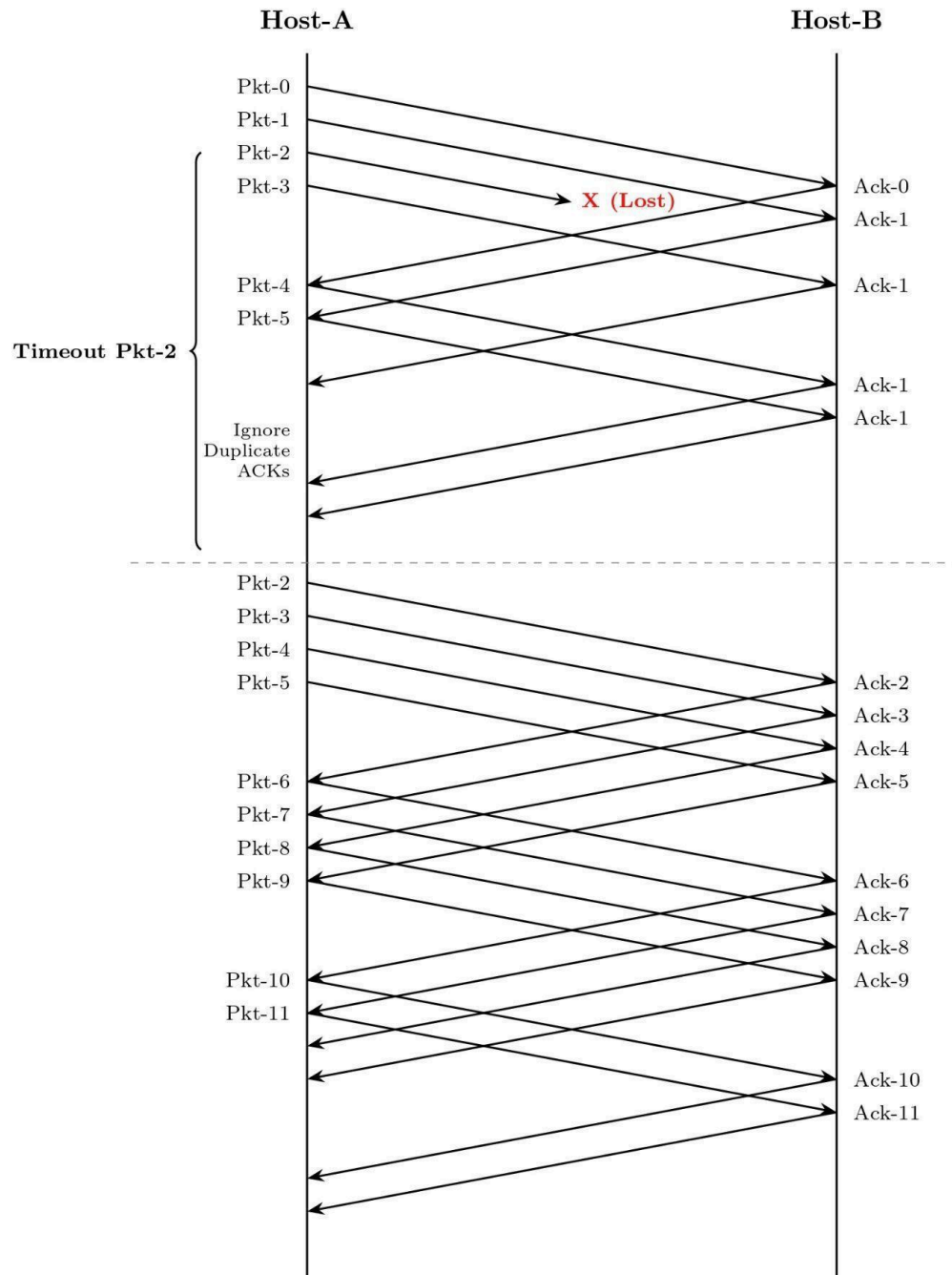


Figure 1: Go-Back-N Protocol Timing Diagram

- b. What is the minimum number of packets (window size W) that must be kept in flight to achieve 100% utilization?

Q2. Consider a reliable data transmission scenario between two hosts, A (sender) and B (receiver). Host A has **12 packets** to transmit and employs a Go-Back-N protocol with **window size $N=4$** and sequence numbers ranging from 0 to 15. If acknowledgements sent by host B were never lost, and only the **3rd packet (having sequence number 2)** was lost, how many packets in total were transmitted by host A? Support your answer through a timing diagram. **[10 Marks] [CLO 2]**

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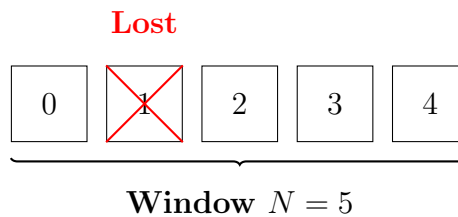
Quiz 3: Transport Layer (Chapter 3)

Name: _____ Roll No: _____ Section: BSE-6B1 (Spring 2026)

1. (5 points) Go-Back-N vs Selective Repeat Analysis

Consider a pipelined protocol over a link with a very high Bandwidth-Delay Product.

- Sender sends packets 0, 1, 2, 3, 4. Packet 1 is lost. All other packets and ACKs arrive correctly. Timeout occurs after Packet 4 is sent. Window Size $N = 5$. Sequence Numbers: 0, 1, 2, ...
- (a) If using **Go-Back-N**, mathematically calculate the total number of packet transmissions (initial + retransmissions) required to successfully deliver packets 0 through 4. If using **Selective Repeat**, calculate the same.
- (b) Will **Selective Repeat** operate correctly if $N = 5$ and $k = 3$? Consider the SR window size constraint, create a scenario where it fails and explain it.



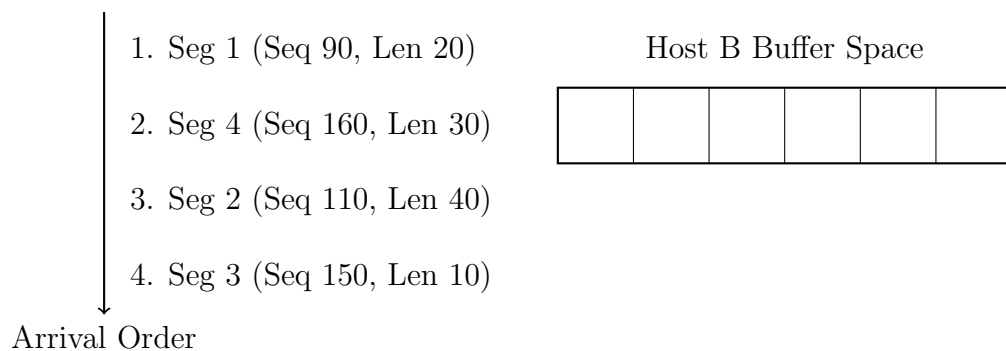
2. (10 points) TCP Sequence Numbers & Re-ordering

Host A sends 4 segments to Host B over a TCP connection.

- Segment 1: Seq = 90, Length = 20 bytes.
- Segment 2: Seq = 110, Length = 40 bytes.
- Segment 3: Seq = 150, Length = 10 bytes.
- Segment 4: Seq = 160, Length = 30 bytes.

Assume the initial Sequence Number is 90. Host B receives them in this order: **Segment 1, Segment 4, Segment 2, Segment 3.**

- (a) List the ACK number Host B sends immediately after receiving EACH segment in that specific order.
- (b) Draw the receive buffer state after Segment 4 arrives but before Segment 2 arrives.



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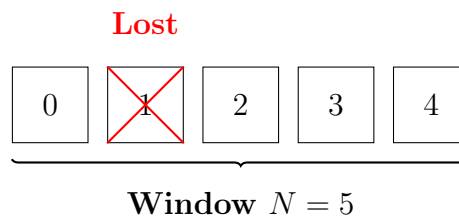
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Name: _____ Roll No: _____ Section: BSE-6B1 (Spring 2026)

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Solution:

(a)

1. **Go-Back-N:** 1. Sender sends 0, 1, 2, 3, 4. (5 transmissions). 2. Receiver gets 0 (ACKs 0). Receiver expects 1, gets 2, 3, 4 (discards them, re-ACKs 0). 3. Sender Timeout for Packet 1. 4. GBN retransmits from base: Sends 1, 2, 3, 4. (4 transmissions). Total transmissions = $5 + 4 = 9$.
2. **Selective Repeat:** 1. Sender sends 0, 1, 2, 3, 4. (5 transmissions). 2. Receiver gets 0 (ACKs 0). Receiver misses 1. Receiver gets 2, 3, 4 (Buffers them, ACKs 2, 3, 4 selectively). 3. Sender Timeout for Packet 1. 4. Sender retransmits ONLY 1. (1 transmission). Total transmissions = $5 + 1 = 6$.

(b)

SR Protocol Constraint: Window Size N must satisfy $N \leq \frac{1}{2} \times 2^k$.

Here $2^k = 8$. Max N should be 4.

Given $N = 5$. Problem:

1. Sender sends 0,1,2,3,4.
2. Rx receives all, sends ACKs, moves window to [5,6,7,0,1].
3. All ACKs lost.

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5. Receiver sees Packet 0. Is this a retransmission of old 0 or new 0 (in window [5,6,7,0,1])? It cannot distinguish. Protocol fails.

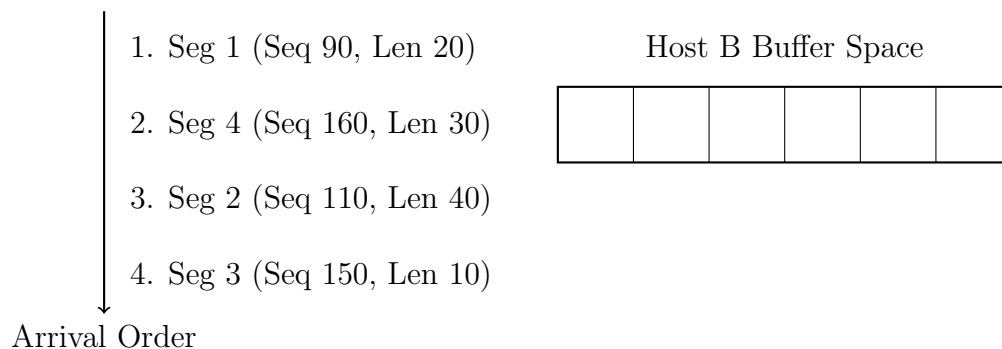
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Solution:

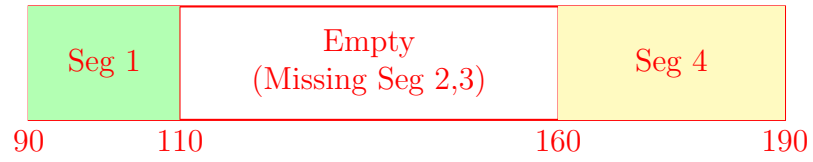
(a) ACK Generation:

1. Rx Seg 1 (90-109): Data is in order. Next expected is 110. **ACK = 110.**
2. Rx Seg 4 (160-189): Gap detected (missing 110-159). Buffer Seg 4. Send duplicate ACK. **ACK = 110.**
3. Rx Seg 2 (110-149): Gap partially filled. Still missing 150. Buffer Seg 2. Send duplicate ACK. **ACK = 150.**
4. Rx Seg 3 (150-159): Gap filled. We have 90 through 189 continuous bytes. Next expected is 190. **ACK = 190.**

(b) Receive Buffer State (After Seg 4, Before Seg 2):

State Description: Segment 1 is delivered/acknowledged. Segment 4 is buffered (out of order). There is a "Hole" where Segment 2 and 3 should be.

Host B Buffer Content (Seq 90 to 190)



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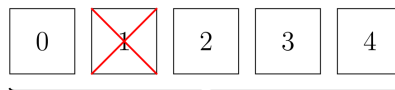
Q1. Carefully read the Question before attempting:

[5 marks]

Consider a pipelined protocol over a link with a very high Bandwidth-Delay Product. The sender sends packets 0, 1, 2, 3, 4. Packet 1 is lost. All other packets and ACKs arrive correctly. Timeout occurs after Packet 4 is sent. Window Size $N = 5$. Sequence Numbers: 0, 1, 2, ...

- If using **Go-Back-N**, mathematically calculate the total number of packet transmissions (initial + retransmissions) required to successfully deliver packets 0 through 4. If using **Selective Repeat**, calculate the same.
- Will **Selective Repeat** operate correctly if $N = 5$ and $k = 3$? Consider the SR window size constraint, create a scenario where it fails and explain it (why)?

Lost



Window $N = 5$

Q2. Carefully read the Question before attempting:

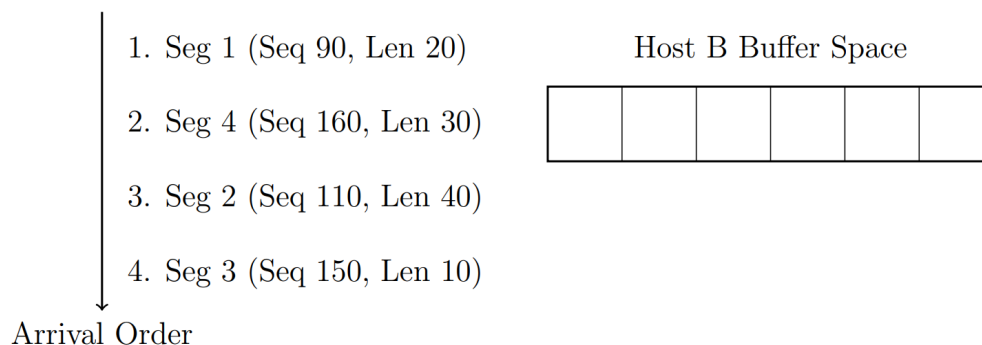
[10 marks]

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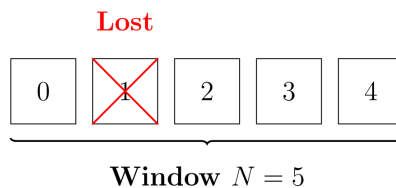
Section:

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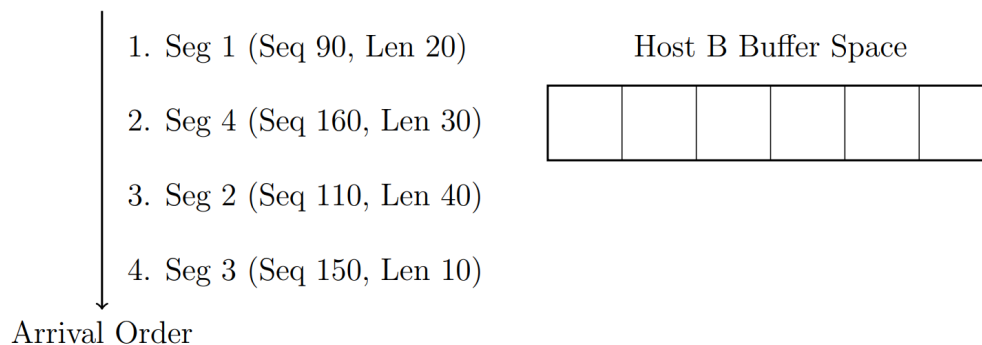
Q2. Carefully read the Question before attempting:**[10 marks]**

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